Jason (Junguang) He

+1(773)273-0966 | junguanghe2023@u.northwestern.edu | LinkedIn

EDUCATION

Ph.D. in Theoretical Physics, Northwestern University 2018 - 2023

B.S. in Physics, University of Science and Technology of China 2014 - 2018

EXPERIENCE

Software Engineer [C++, Python, Java, Kotlin] Nov 2025 - Present
Google Sunnyvale, CA

- **Firestore Watch:** Implemented support for 16MB documents in Firestore Watch, the real-time engine to stream live database changes. Designed and ran stress tests to identify and fix potential issues and bottlenecks.
- **Client Proxy:** Optimized the Client Proxy of Firestore Watch by refining the memory tracking, resource management, and implementing tenant-aware egress throttling, which significantly improved the system stability.
- **Subscription Server:** Implemented, verified and optimized the Sharded Reverse Query Matcher (RQM) in Firestore Watch, which improves efficiency and reduces query matching overhead.
- **Shredded Columnar:** Implemented observability and controllability for shredded columnar in Firestore, which achieves 200x speedup for specific query shapes.
- **GCE Prober:** Designed, implemented and deployed continuous external prober in Google Compute Engine (GCE) for Cloud Firestore to monitor performance and stability, ensuring high availability of the service and avoiding potential outages.

Software Engineer [C++, Python] May 2025 - Sep 2025
Tensor Auto San Jose, CA

- **Xsim Platform:** Took ownership of the autonomous vehicle simulation platform, establishing capabilities for ongoing support, maintenance, and future development.
- **Challenge Injection:** Designed and implemented a rule-based challenge injection system within the offboard simulation pipeline, enabling configurable injection of diverse challenge scenarios.

Quantitative Developer [C++, CUDA, Python] Jan 2023 - Apr 2025
Aleto, Magma Capital Management Chicago, IL

- **cuGP:** Used CUDA to develop a GPU-based feature-generating framework based on genetic programming and symbolic transformation. Achieved more than 100x speedup compared with gplearn. Implemented customized cross-section operators and time-series operators with rolling boundaries.
- **GPU Optimization:** Used Cooperative Groups to achieve grid-level synchronization and reduce the number of kernel launches. Achieved memory coalesced access and avoided warp divergence to optimize GPU performance.
- **Trading System:** Designed and implemented a low-latency trading system consisting of a market data consumer, trade engine, and order gateway connected by lock-free queues. Also implemented a data replay server for simulation and debug purposes, using UDP and TCP sockets to communicate with the trading system.
- **E2E Pipeline:** Led a team in constructing a comprehensive interday cross-sectional strategy pipeline from the ground up, covering historical data processing, feature engineering, modeling, signal generation, risk management, backtesting, paper trading, live execution and trading analysis.
- **Alpha Research:** Researched and developed alpha signals across multiple categories, including momentum, liquidity, volatility, cyclicity, correlation, and distribution. Designed and tested advanced feature combination techniques to enhance trading signal generation.

Machine Learning Engineer Intern [Python, C++, SQL] Jun 2022 - Sep 2022
Meta Menlo Park, CA

- **Watch Ranking:** Worked on Facebook Watch recommendation ranking, focusing on cold-start users.